

Maintenance procedures

PLTMG DUR1 - P/12032

System

7xW18V50DF

Engine No

PAAE237323- 9

Running hour based jobs

<u>Job duty</u>	<u>Interval</u>	<u>Num jobs</u> <u>Wärtsilä</u>	<u>Num jobs</u> <u>Customer</u>
Daily routine checks	24		666
Water cleaning of compressor	50		320
Check water quality	500		32
Take oil sample	500		32
Measure the pressure drop over charge air cooler(s)	500		32
Water cleaning of turbine	500		32
Maintenance of control mechanism	500		24
Function check of oil mist detector	500		32
Check tightening of engine fastening bolts	1 000		1
Replace pilot fuel oil filter cartridges	1 000		16
Check electropneumatic overspeed trip device	2 000		8
Functional check of control systems	2 000		8
Change oil in the governor	2 000		8
Check yoke and valve clearances	2 000		8
Visual inspection of valve rotators	2 000		8
Make the leak test (gas system)	2 000		8
Check control mechanism	2 000		8
Change fresh air filter in oil mist detector	2 000		8
Check connectors and cables	4 000		4
Check crankshaft alignment	4 000		4
Inspect contact faces of camshaft	4 000		4
Check the alingment (flexible mounted engine)	4 000		4
Check the wastegate valve and actuator	4 000		3
Clean the charge air cooler	4 000		4
Check function of mechanical overspeed trip device	6 000		2
Inspect flexible pipe connections	6 000		1
Inspect exhaust duct supports	6 000		2
Visual inspection of expansion bellows	6 000		2
Inspect injection valves, change nozzles	6 000		2
Check adjustment of the pressure control valve	8 000		2
Clean the air filter	12 000		1
Grease the secondary shaft of the turning gear	12 000		1
Renew flexible pipe connections	12 000		1
Overhaul of injection pumps	12 000		1
Dismount and clean turbocharger	12 000		1
Inspect (replace) turbocharger bearings	12 000		1
General overhaul of wastegate valve and actuator	12 000		1

DISCLAIMER

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Spare parts per year

PLTMG DURI - P/12032

System

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Year1

Running hours: 0 to 8000

Part number	Description	Quantity
155076	Gasket set, complete	28
167046	Sealing set	126
167167	Nozzle	126
2071320	Sealing ring	28
207926	Pilot valve	14
354414	O-ring	56
3551756	O-ring	56
3551760	O-ring	56
356420	Fresh air filter	28
470130	Filter cartridge	112
470447	Sealing set for fuel filter	7
Number of different parts		11

Year2

Running hours: 8001 to 16000

Part number	Description	Quantity
155076	Gasket set, complete	28
156379	Spare part set	14
156380	Spare part set	14
161016	O-ring	126
161036	O-ring	126
165020	Sealing ring	252
165021	Core plug	252
165178	Sealing set for pump overhaul	126
165202	Sealing ring	378
167046	Sealing set	126
167167	Nozzle	126
2071107	Sealing set for actuator	7
2071320	Sealing ring	28
207926	Pilot valve	14
207997	Sealing set, butterfly valve	7
350107	O-ring	252
350109	O-ring	126
350110	Support ring	252
350156	Guide ring	126
354414	O-ring	56
3551756	O-ring	56
3551760	O-ring	56
356420	Fresh air filter	28
470130	Filter cartridge	112
470447	Sealing set for fuel filter	7
FH101	Flexible hose, Fuel inlet	7
FH102	Flexible hose, Fuel outlet	7
FH108	Flexible hose, Gas inlet	7
Number of different parts		28

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Other parts

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Safety spare parts

<u>Part number</u>	<u>Description</u>	<u>Quantity</u>
100037	Nut	12
100096	Main bearing shell (upper)	6
100097	Main bearing shell (lower)	6
100100	Cylinder liner	3
100101	Antipolishing ring	3
100109	O-ring	3
100111	O-ring	6
100112	Thrust bearing half	12
100113	Bearing shell	6
100114	Bearing bush	6
100163	Stud	12
111004	Connecting rod, lower part	3
111006	Nut	6
111007	Screw	6
111008	Screw	6
111009	Nut	6
111016	Bearing shell, upper	6
111017	Bearing shell, lower	6
111018	Shim	3
111024	Connecting rod, upper part	3
113010	Securing ring	6
113013	Piston ring set	6
113014	Gudgeon pin	3
113025	Piston, compl.	3
120012	Seat ring, inlet	12
120025	Yoke pin	6
120055	Cylinder head with valves	3
120070	Sealing set for cylinder head	54
120080	Repair kit for cylinder head overhaul	54
120081	Valve guide	12
120103	Seat ring, exhaust	12
121002	Valve cotter	24
121004	O-ring	24
121007	Inlet valve	12
121008	Valve spring, outer	12
121009	Valve spring, inner	12
121010	Exhaust valve	12
121030	Valve rotator, compl.	24
123004	Slide-o-ring	6

123008	Slide-o-ring	6
123010	O-ring	2
123015	Starting valve, compl.	3
123025	Sealing ring	6
123461	O-ring	6
123463	O-ring	6
123464	O-ring	6
124022	O-ring	12
124028	Screw	9
124036	Clamping ring, complete	3
124037	Clamping ring, complete	3
124044	Sealing ring	3
141014	Bearing bush	6
141025	Valve tappet	6
141030	Tappet roller	6
141031	Locking pin	6
141032	Roller pin	6
141036	Compression spring	6
141037	Valve tappet, complete	6
143015	Yoke, compl., Inlet Valves	3
143016	Yoke, compl., Outlet valves	3
143036	Screw	12
143037	O-ring	3
143039	Rocker arms and bracket	3
148016	Stud (alt. 148188)	90
148035	Snap ring	6
148036	Guiding pin	6
148097	Camshaft piece	3
148099	Camshaft piece	3
1561071	Spare part set	3
1561072	Spare part set	3
161002	Injection pump tappet	3
161003	Tappet roller	3
161004	Roller pin	3
161005	Compression spring	3
161006	Locking pin	3
161007	Adjusting screw	3
161008	Screw	3
161009	Securing plate	3
161016	O-ring	3
161021	Compression spring	3
161036	O-ring	3
161039	Spring plate	3
161040	Tappet, complete	3
164006	O-ring	12
164010	Main gas admission valve	6
164026	O-ring	12
165006	Spring	3

165026	Spring	3
165102	Pump element	54
165108	Cover, complete	3
165124	Spring plate, upper	3
165125	Control sleeve	3
165132	Spring plate, lower	3
165133	Retainer ring	3
165172	Rack	3
165174	Shot, tappet	3
165178	Sealing set for pump overhaul	54
165180	Injection pump, compl.	3
165207	Flange, complete	3
165210	Retainer ring	3
167001	Injection valve, compl.	54
167035	Connection piece	3
167046	Sealing set	54
167150	Main injection pipe, compl.	3
167167	Nozzle	54
167409	Injection pipe	6
174127	Pilot fuel pump	3
174234	Control valve, complete	3
191235	Sealing set	3
191236	Sealing set	3
200252	Bellows	6
200327	Nut	72
200328	Bellows	6
200329	Screw	18
200333	Gasket	12
200338	Screw	144
200339	Nut	144
200340	Sealing ring	12
200369	Screw	72
200378	Bellows	6
207184	Butterfly valve + positioner	3
220001	Speed governor	3
228158	Solenoid valve	3
3501055	Fuel pipe	6
357044	O-ring	6
357712	Converter	3
470130	Filter cartridge	6
5062606	Knock sensor	3
CCM20-A1-3	Control module	3
CCM20-B1-3	Control module	3
GS171	Limit switch	3
LS204	Level switch	3
PT101	Pressure sensor	3
PT125	Pressure transmitter	3
PT201	Pressure sensor	3

PT301	Pressure sensor	3
PT401	Pressure transmitter	3
PT5011A - 5091A	Cylinder pressure sensor	3
PT601	Pressure sensor	3
PT700	Pressure transmitter	3
ST196P	Speed sensor	3
ST197P	Limit switch	3
TE201	Temperature sensor	3
TE5011A - 5091A	Exhaust gas temperature sensor	3
TE700	Temperature sensor	3
TE701	Temperature sensor	3
TE7011A - 7091A	Temperature sensor	3
Number of different parts		147

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